

Introduction to Parallel Algorithms and Programming

Session 2: Parallel algorithms in the PRAM model (part 1)

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Objectives

- Revise sequential algorithms, the **RAM** model, and complexity analysis
- Introduce the **PRAM** model of parallel computation and its variants: EREW, CREW, CRCW
- Learn what a PU knows, and what a valid PRAM algorithm looks like on each variant
- Design and analyse first PRAM algorithms: array multiplication, search, broadcast, reduction
- Learn the **pointer jumping** technique: list ranking, prefix computation on a list
- Learn the metrics for the complexity and performance of PRAM algorithms: time, work, cost, speedup, efficiency
- Apply them to every algorithm of the session: which ones are work-optimal, which ones are efficient?

Outline

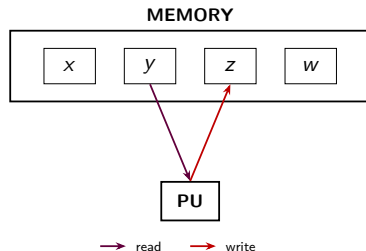
- 1 The RAM model and sequential complexity
- 2 The PRAM model
- 3 A first PRAM algorithm: multiplying an array by 2
- 4 Search in an array
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The RAM model

The RAM (random access machine) model is an abstraction of a sequential computer.

- Consists of a processing unit (PU) and an associated memory space.
- Access to each element in the memory is done in constant time.
- Each operation is performed in constant time in the PU.
- Equivalent to a Turing machine (simulation); belongs to the class of “register machines”.
- Not very realistic (no L1/L2/L3/DRAM memory hierarchy), yet useful for developing algorithms that are **asymptotically optimal**.
 - One should then adapt these algorithms to modern computer architectures (i.e., multicore CPUs, GPUs, etc., the focus of HPC).



Algorithms and complexity

An algorithm is a sequence of operations on a RAM machine, which aims to solve a given problem.

- Each memory access and arithmetic operation takes unit time.
- The **time complexity** of the algorithm is a function $T(N)$ of the problem size, which represents the total number of operations it performs.
- One can also consider its **space complexity**, the total memory space it uses in execution.

SUMARRAY($A[1 \dots N]$)

```
1:  $sum \leftarrow 0$   
2: for  $i = 1, \dots, N$  do  
3:    $sum \leftarrow sum + A[i]$   
4: return  $sum$ 
```

Complexity analysis of SUMARRAY

Count the elementary operations (memory accesses, arithmetic, comparisons), each taking unit time.

- Line 1: one write, 1 operation.
- Lines 2–3: N iterations; each iteration reads $A[i]$ and sum , adds, writes sum , increments and tests i : a constant number c of operations.
- $T(N) = cN + O(1) = O(N)$: **linear** time.
- The exact constant c depends on the machine and the compiler; only the **order of growth** $O(N)$ is meaningful, algorithmically speaking.
- Space complexity: $O(N)$. The input itself must be stored; we cannot ignore the input size.

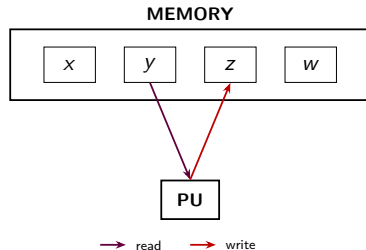
SUMARRAY($A[1 \dots N]$)

```
1:  $sum \leftarrow 0$ 
2: for  $i = 1, \dots, N$  do
3:    $sum \leftarrow sum + A[i]$ 
4: return  $sum$ 
```

Complexity analysis

Many methods exist for complexity analysis:

- Nested loops: counting techniques in the loop iteration domain
- Recursion / divide-and-conquer: counting by hand, induction, Master theorem
- Probabilistic analysis (e.g., quicksort)
- Amortized analysis (e.g., dynamic tables)
- These analyses can be performed for
 - the best case
 - the worst case
 - the average case

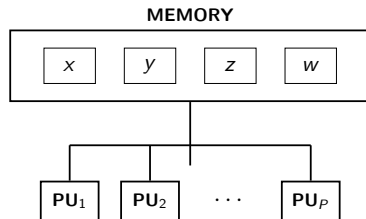


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The PRAM model

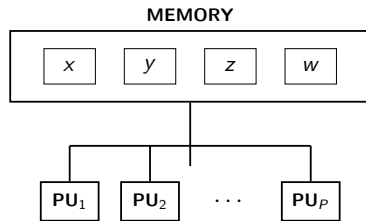
The parallel random access machine (PRAM) is an abstraction of a parallel computer.

- Consists of P PUs and a memory space shared among all PUs.
- PUs
 - execute the same algorithm/code in a **synchronous manner**, but on different elements in the memory (i.e., à la SIMD): all of them attempt to execute the **same instruction** of the code in the **same clock cycle**;
 - more akin to a synchronous **SIMT** execution (rather than SIMD, which operates on contiguous data elements), as in GPUs (which are only partially synchronous);
 - can perform an arithmetic operation, or read from / write to different memory locations, **simultaneously** in constant time;
 - can have **private memory/registers** for PU-specific information (identifier, program counter) and temporary results (e.g., data loaded from the shared memory before an arithmetic operation).



The PRAM model (cont.)

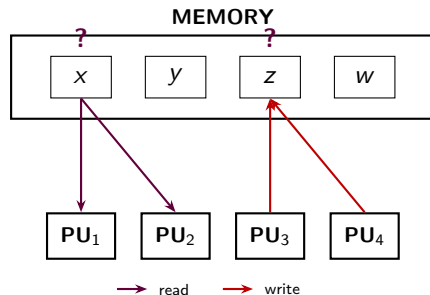
- Read/write on the same memory location?
 - This behaviour defines the different PRAM variants.
- Number of PUs: potentially infinite!
- Memory size: potentially infinite!
- Communication cost: completely ignored (done implicitly through the fast shared memory)
 - PRAM is useful for developing **asymptotically optimal** parallel algorithms.
 - One should then adapt these algorithms to modern parallel computer architectures using HPC techniques.



Memory conflicts in PRAM

During the execution of a PRAM algorithm, multiple PUs could try to access the same memory location (memory conflict).

- Possible scenarios?
 - **Concurrent reads** of the same memory location by multiple PUs.
 - **Concurrent writes** to the same memory location by multiple PUs.
- Each of them is either allowed or forbidden by the model.
- This yields four PRAM variants: **CREW**, **CRCW**, **EREW**, and ERCW.

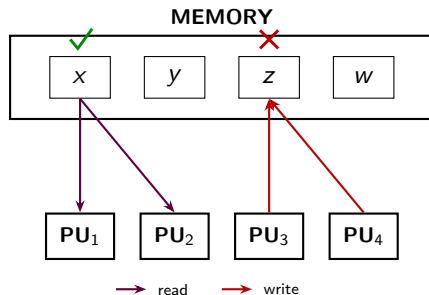


PU₁, PU₂ read x at the same time; PU₃, PU₄ write z at the same time. Allowed?

The CREW PRAM model

- **CREW: Concurrent Read, Exclusive Write**

- Simultaneous reading of the same memory location is allowed, and takes constant time.
- Simultaneous writing is forbidden (bug).
- The standard PRAM model in the literature.

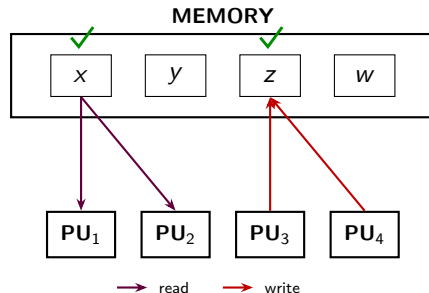


Concurrent read of *x*: **allowed**. Concurrent write of *z*: **forbidden**.

The CRCW PRAM model

- **CRCW: Concurrent Read, Concurrent Write**

- Simultaneous reading/writing from/to the same memory location is allowed, and takes constant time.
- The most powerful model.
- A “concurrency mode” is employed for simultaneous writes:
 - **Consistent** mode: all PUs write the same value.
 - **Arbitrary** mode: one PU's value is retained, chosen arbitrarily.
 - **Priority** mode: the value of the PU with the lowest index is retained.
 - **Fusion** mode: a commutative and associative operation is applied (+, ×, min, max, and, or, xor, ...).
- Sometimes the mode is imposed by the problem; other times we have the liberty to choose the mode to employ.
- Not very realistic: real hardware has a hard time handling simultaneous write requests to the same memory location (mechanisms exist, e.g., atomic operations, but they are quite expensive).

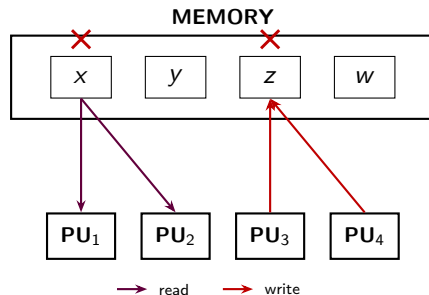


Concurrent read of x: **allowed**. Concurrent write of z: **allowed**, resolved by the concurrency mode.

The EREW PRAM model

- **EREW: Exclusive Read, Exclusive Write**

- Simultaneous read/write of the same memory location by multiple PUs is forbidden.
- The most restrictive model.

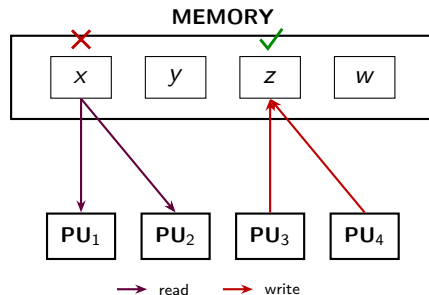


Concurrent read of *x*: **forbidden**. Concurrent write of *z*: **forbidden**. Only accesses to different cells may happen at the same time.

The ERCW PRAM model

- **ERCW: Exclusive Read, Concurrent Write**

- Does not exist. Why?
 - Does not make sense architecturally: hardware that can merge concurrent writes can serve concurrent reads.



Concurrent read of x : **forbidden**, but
concurrent write of z : **allowed**?!

Which model is the most realistic: CREW or EREW?

Both are, depending on the **scope** (“granularity”) of the hardware we look at.

Small scope: CREW-like behaviour

- **SIMD lanes:** one set(`val`, `vector`) instruction copies `val` into all 8 lanes of `vector` in one cycle.
- **Cores of one CPU:** each core keeps a copy of what it reads in its cache; once loaded, many cores read the same value for free.
- **GPU threads:** reading the same address, the hardware fetches it once and hands it to all of them.
- Concurrent *writes* to one address are always serialized: exclusive write.

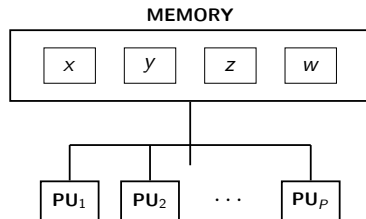
Large scope: EREW-like behaviour

- The *first* time many cores need a value stored far away, the requests queue up and are served one by one.
- **Machines of a cluster:** a value lives in one machine’s memory; any other machine needing it must receive a message. Reading is a communication; sending one value to P machines takes $\log_2 P$ rounds.

No need to pick a side: the smaller the scope of the hardware, the more CREW-like its behaviour; the larger the scope, the more EREW-like.

Writing PRAM algorithms: what does a PU know?

- In the **private space** (registers) of each PU:
 - its index i , and the number of active PUs in the parallel region, say N ;
 - a **copy of the program**: reading an instruction never causes a read conflict, even on an EREW PRAM;
 - the **array pointers** (addresses of A , $next$, ...) and the constants of the program, considered part of the program: each PU has its own private copy;
 - temporary values, handled transparently by the registers (we will not write them):
 - a read of a shared memory cell is a **load**: it puts the value into a private register;
 - the arithmetic operation is then performed on the **private registers**;
 - an assignment to a shared memory cell is a **store**: the value of a private register is written into the cell.
- The **real data** lives in the **shared memory**: $A[1 \dots N]$, $next[1 \dots N]$, ...
 - All PUs, each with its own index i , accessing the element $A[i]$ at the same time = accessing the cells (address of A) + i : different PUs, different cells, **exclusive** access.
 - Several PUs reading the *same* cell in the same step:
concurrent read. Fine on CREW/CRCW, forbidden on EREW.



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A first PRAM algorithm: multiplying an array by 2

Given an array $A[1 \dots N]$, multiply each element by 2.

- Sequential algorithm/complexity?
 - One loop over the array: $O(N)$.
- PRAM algorithm: one PU per element, N PUs.
Line 2 = three elementary steps for PU i :
 - $temp \leftarrow \text{LOAD}(A + i)$: the value $A[i]$ goes into a private register;
 - $temp \leftarrow 2 \cdot temp$: arithmetic on the register (2 is a constant of the program);
 - $\text{STORE}(A + i, temp)$: the register is written back into $A[i]$.
 - Done by all PUs **simultaneously**: parallel time complexity $T(N) = O(1)$, whatever N !
- Which PRAM models can run it?
 - Reads: PU i reads only cell $A[i]$: **exclusive**.
 - Writes: PU i writes only cell $A[i]$: **exclusive**.
 - Runs on an **EREW** PRAM, hence also on CREW and CRCW.

MULTIPLYBYTWO($A[1 \dots N]$)

```
1: forall  $i = 1, \dots, N$  in parallel do
2:    $A[i] \leftarrow 2 \cdot A[i]$ 
```

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Another PRAM algorithm: search in an array

Given an array $A[1 \dots N]$ and a key e , find an index i such that $A[i] = e$.

- Sequential: $O(N)$. PRAM: $O(1)$ with N PUs, all comparisons at once.
- Which PRAM models can run it? **Reads** first (line 3): $A[i]$ is exclusive, no problem; e is read by all PUs at the same time: **not exclusive**, EREW does not work.
- **Writes**: $idx \leftarrow 0$ (line 1) is in the sequential part; $idx \leftarrow i$ (line 4) is in the parallel part. Can this run on a CREW PRAM? Nuanced: if the elements of A are **unique**, at most one PU writes idx : valid CREW algorithm. Otherwise, several PUs may write idx simultaneously: **CRCW** required.
- CRCW: which concurrency modes would work?
 - Consistent? No: different values i are written.
 - Arbitrary? Yes: one of the matching indices.
 - Priority (lowest PU index)? Yes: the first occurrence.
 - Fusion with +? No: a sum of indices. With min/max? Yes: the first/last occurrence.

SEARCHINDEX($A[1 \dots N], e$)

```
1:  $idx \leftarrow 0$ 
2: forall  $i = 1, \dots, N$  in parallel do
3:   if  $A[i] = e$  then
4:      $idx \leftarrow i$ 
5: return  $idx$ 
```

Search on an EREW PRAM: how?

The only obstacle for EREW: all PUs read e at the same time (line 3).

- Solutions?

- Give each PU its **own copy** of e , in the shared memory: an array $pe[1 \dots N]$ with $pe[i] = e$.
- Line 3 becomes **if** $A[i] = pe[i]$: PU i reads only $pe[i]$, an **exclusive** read.
- How to produce $pe[1 \dots N]$ from e , without any concurrent read?
 - Making N copies of one value is the **broadcast** operation.

SEARCHINDEX($A[1 \dots N], e$)

```
1:  $idx \leftarrow 0$ 
2: forall  $i = 1, \dots, N$  in parallel do
3:   if  $A[i] = e$  then
4:      $idx \leftarrow i$ 
5: return  $idx$ 
```

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Broadcast on an EREW PRAM

Given a value e , produce an array $pe[1 \dots N]$ with $pe[i] = e$ for all i , without any concurrent read.

- Naive: PU 1 copies e into $pe[1], pe[2], \dots, pe[N]$ one by one: $O(N)$, no better than the sequential algorithm!
- **Doubling**: every PU that already has the value passes it on to one PU that does not: 1, 2, 4, 8, ... copies.
- Step k : PU $i \leq 2^k$ reads $pe[i]$ and writes $pe[i + 2^k]$: all accesses are **exclusive**.
- Parallel time: $\lceil \log_2 N \rceil$ iterations of an $O(1)$ body: $T(N) = O(\log N)$ with N PUs.
- The fundamental EREW building block: needed for any value that is not in the program.

BCAST(e, N)

```
1:  $pe[1] \leftarrow e$ 
2: for  $k = 0, \dots, \lceil \log_2 N \rceil - 1$  do
3:   forall  $i = 1, \dots, N$  in parallel do
4:     if  $i \leq 2^k$  and  $i + 2^k \leq N$  then
5:        $pe[i + 2^k] \leftarrow pe[i]$ 
6: return  $pe[1 \dots N]$ 
```

Search on an EREW PRAM

Given an array $A[1 \dots N]$ and a key e , find an index i such that $A[i] = e$ (unique elements).

- First broadcast e into $pe[1 \dots N]$, then PU i compares $A[i]$ with $pe[i]$.
- All accesses are now exclusive: runs on an **EREW** PRAM.
- Complexity with N PUs?
 - Line 2: $O(\log N)$ (BCAST).
 - Lines 3–5: $O(1)$.
 - $T(N) = O(\log N)$, instead of $O(1)$ on a CREW PRAM.
- The whole $\log N$ comes from the broadcast: the price of exclusive reads.

SEARCHINDEXEREW($A[1 \dots N], e$)

```
1:  $idx \leftarrow 0$ 
2:  $pe[1 \dots N] \leftarrow \text{BCAST}(e, N)$ 
3: forall  $i = 1, \dots, N$  in parallel do
4:   if  $A[i] = pe[i]$  then
5:      $idx \leftarrow i$ 
6: return  $idx$ 
```

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Back to SUMARRAY: on a CRCW PRAM

Given an array $A[1 \dots N]$, compute $A[1] + A[2] + \dots + A[N]$. Sequential: $O(N)$.

- How to do it on a CRCW PRAM with N PUs?
 - All PUs write their element into the *same* cell *sum*: a concurrent write.
 - Choose the **fusion mode with +**: the N written values are added, and *sum* receives their total.
 - The mode is *our* choice here: the problem does not impose it.
- Complexity?
 - One read of $A[i]$ and one (concurrent) write: $T(N) = O(1)$ with N PUs.
- The whole work of summing is done by the memory, “for free”. Which is exactly why CRCW is not very realistic.

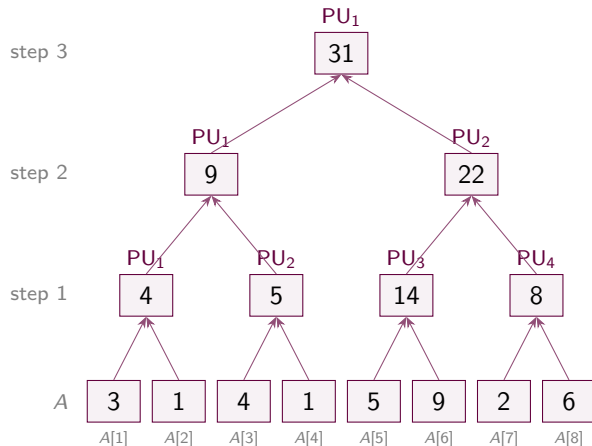
SUMARRAYCRCW($A[1 \dots N]$)

```
1:  $sum \leftarrow 0$ 
2: forall  $i = 1, \dots, N$  in parallel do
3:    $sum \leftarrow A[i]$            ► fusion +
4: return  $sum$ 
```

SUMARRAY on a CREW/EREW PRAM

Given an array $A[1 \dots N]$, compute $A[1] + A[2] + \dots + A[N]$ without any concurrent write.

- No fusion mode: a PU can add only two values at a time.
- Idea: add the elements **two by two**, in parallel: $A[1] + A[2]$, $A[3] + A[4]$, ...
- The array size is **halved** at each step; repeat until one element is left.
- $N = 8$: three steps, with 4, 2, then 1 PUs active.
- In general: $\lceil \log_2 N \rceil$ steps.



SUMARRAY on a CREW/EREW PRAM (cont.)

From the tree to PRAM code: at each step, PU i produces the i -th element of the next level.

- PU i reads $A[2i - 1]$ and $A[2i]$ (only $A[2i - 1]$ if n is odd), writes $A[i]$: each cell is read by one PU and written by one PU: **EREW**.
 - PU i writes $A[i]$ while PU $\lceil i/2 \rceil$ reads it, in the same line: fine, reads happen before writes.
- Complexity: the loop body (lines 3–8) is $O(1)$; n is halved at each iteration: $\lceil \log_2 N \rceil$ iterations, known in advance (n lives in the private space of each PU, no termination test).
- $T(N) = O(\log N)$ with $\lceil N/2 \rceil$ PUs. The input array is overwritten.

SUMARRAYEREW($A[1 \dots N]$)

```
1:  $n \leftarrow N$ 
2: while  $n > 1$  do
3:   forall  $i = 1, \dots, \lceil n/2 \rceil$  in parallel do
4:     if  $2i \leq n$  then
5:        $A[i] \leftarrow A[2i - 1] + A[2i]$ 
6:     else
7:        $A[i] \leftarrow A[2i - 1]$ 
8:    $n \leftarrow \lceil n/2 \rceil$ 
9: return  $A[1]$ 
```

Generalizing: reduction on an EREW PRAM

Given $A[1 \dots N]$ and a binary operation $\otimes$, compute $A[1] \otimes A[2] \otimes \dots \otimes A[N]$. Same idea, written **recursively**: combine two by two, then reduce the half-size array.

- Which operations $\otimes$? The pairs are combined in a different order than the sequential left-to-right computation: $\otimes$ must be **associative**, so that regrouping does not change the result. Commutativity is *not* needed: adjacent pairs $(2i - 1, 2i)$ keep the order (it would be if PU i combined $A[i]$ with $A[i + \lceil n/2 \rceil]$).
- Interesting $\otimes$: $+$, $\times$, $\min$, $\max$, logical/bitwise and, or, xor, gcd, matrix product, ...
- Complexity: $T(n) = T(\lceil n/2 \rceil) + O(1)$, hence $T(N) = O(\log N)$ with $\lceil N/2 \rceil$ PUs.
- **Reduction** is the **mirror image of broadcast**: BCAST turns 1 value into N copies by doubling, REDUCE turns N values into 1 by halving; $\lceil \log_2 N \rceil$ steps each.

REDUCE($A[1 \dots n], \otimes$)

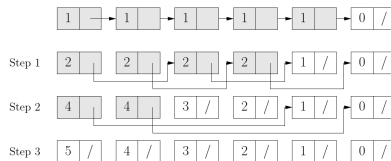
```
1: if  $n = 1$  then return  $A[1]$ 
2: forall  $i = 1, \dots, \lceil n/2 \rceil$  in parallel do
3:   if  $2i \leq n$  then
4:      $A[i] \leftarrow A[2i - 1] \otimes A[2i]$ 
5:   else
6:      $A[i] \leftarrow A[2i - 1]$ 
7: return REDUCE( $A[1 \dots \lceil n/2 \rceil], \otimes$ )
```

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List ranking (cont.)

How to parallelize in PRAM?

- Idea: **recursive pointer jumping**
 - First, each pointer jump adds 1 to the distance.
 - Then, as we jump over pointers, we add the distance of the pointer we jump over.
 - Continue until all pointers point to the end of the list.
 - How many iterations until convergence?
 - At each jump, either the jump distance of a pointer doubles, or it points to the end of the list.
 - In $\log_2 N$ iterations, all pointers point to the end of the list, with the correct distance $d[i]$.



List ranking (cont.)

How to write the PRAM algorithm?

- Initialization (lines 1–2) is done in $O(1)$.
- The main loop terminates after $\log_2 N$ iterations, using N PUs.

RANKCOMPUTATION($next[1 \dots N]$)

```
1: forall  $i = 1, \dots, N$  in parallel do
2:   if  $next[i] = \text{Nil}$  then  $d[i] \leftarrow 0$  else
      $d[i] \leftarrow 1$ 
3:   while there exists a node  $i$  such that
      $next[i] \neq \text{Nil}$  do
4:     forall  $i = 1, \dots, N$  in parallel do
5:       if  $next[i] \neq \text{Nil}$  then
6:          $d[i] \leftarrow d[i] + d[next[i]]$ 
7:          $next[i] \leftarrow next[next[i]]$ 
```

List ranking (cont.)

Read/write conflicts in lines 6–7?

- Line 6 = elementary steps for PU i ($temp1, \dots, temp4$: private registers):
 - $temp1 \leftarrow \text{LOAD}(next + i)$
 - $temp2 \leftarrow \text{LOAD}(d + temp1) \quad \triangleright d[next[i]]$
 - $temp3 \leftarrow \text{LOAD}(d + i) \quad \triangleright d[i]$
 - $temp4 \leftarrow temp3 + temp2$
 - $\text{STORE}(d + i, temp4)$
- Step 2: two PUs never have the same $next$ (injective): **exclusive**. Steps 1, 3, 5: cell i only: **exclusive**.
- Line 7 likewise:
 $temp5 \leftarrow \text{LOAD}(next + temp1)$,
 $\text{STORE}(next + i, temp5)$: exclusive.
- All loads of a line happen before its store: PU i reads the *old* values. Constant number of steps: asymptotically the same! **EREW** is sufficient.

RANKCOMPUTATION($next[1 \dots N]$)

```

1: forall  $i = 1, \dots, N$  in parallel do
2:   if  $next[i] = \text{Nil}$  then  $d[i] \leftarrow 0$  else
      $d[i] \leftarrow 1$ 
3: while there exists a node  $i$  such that
    $next[i] \neq \text{Nil}$  do
4:   forall  $i = 1, \dots, N$  in parallel do
5:     if  $next[i] \neq \text{Nil}$  then
6:        $d[i] \leftarrow d[i] + d[next[i]]$ 
7:        $next[i] \leftarrow next[next[i]]$ 

```

List ranking (cont.)

How to know when to stop (line 3)? On a **CRCW** PRAM: a shared flag *done*, written in **fusion mode “and”**.

- Line 9: every active PU writes ($next[i] = \text{Nil}$) into the same cell *done*: a concurrent write, merged by the “and” fusion mode. *done* is true iff every active node has reached the end.
- Line 4: all PUs read *done* at the same time: a concurrent read, fine on CRCW.
- Inactive PUs (already at the end) do not write; they are done anyway.
- Cost of the test: $O(1)$ per iteration.
 $T(N) = O(\log N)$ with N PUs.

RANKCOMPUTATIONCRCW($next[1 \dots N]$)

```

1: forall  $i = 1, \dots, N$  in parallel do
2:   if  $next[i] = \text{Nil}$  then  $d[i] \leftarrow 0$  else 1
3:  $done \leftarrow \text{false}$ 
4: while not  $done$  do
5:   forall  $i = 1, \dots, N$  in parallel do
6:     if  $next[i] \neq \text{Nil}$  then
7:        $d[i] \leftarrow d[i] + d[next[i]]$ 
8:        $next[i] \leftarrow next[next[i]]$ 
9:        $done \leftarrow (next[i] = \text{Nil})$  ► and
10: return  $d[1 \dots N]$ 

```

List ranking (cont.)

How to know when to stop? On a **CREW/EREW** PRAM: no fusion mode, the flag must be computed by a **reduction**.

- Line 9: each PU writes its own bit $pd[i]$ (exclusive). Line 10: REDUCE with “and”: $O(\log N)$.
- Line 4: all PUs read *done*: fine on CREW. On EREW, add $pdone \leftarrow \text{BCAST}(done, N)$ after line 10 ($O(\log N)$ too) and test $pdone[i]$.
- Complexity: $\lceil \log_2 N \rceil$ iterations, each $O(1)$ for the jump plus $O(\log N)$ for the test: $T(N) = O(\log^2 N)$ instead of $O(\log N)$!
- **Solution**: perform the check only every $\lceil \log_2 N \rceil$ iterations (at most $\lceil \log_2 N \rceil - 1$ useless iterations). The amortized cost of the test is $O(1)$ per iteration, and the overall complexity stays $O(\log N)$.

RANKCOMPUTATIONCREW($next[1 \dots N]$)

```
1: forall  $i = 1, \dots, N$  in parallel do
2:   if  $next[i] = \text{Nil}$  then  $d[i] \leftarrow 0$  else 1
3:  $done \leftarrow \text{false}$ 
4: while not  $done$  do
5:   forall  $i = 1, \dots, N$  in parallel do
6:     if  $next[i] \neq \text{Nil}$  then
7:        $d[i] \leftarrow d[i] + d[next[i]]$ 
8:        $next[i] \leftarrow next[next[i]]$ 
9:        $pd[i] \leftarrow (next[i] = \text{Nil})$ 
10:   $done \leftarrow \text{REDUCE}(pd[1 \dots N], \text{and})$ 
11: return  $d[1 \dots N]$ 
```

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- 2 The PRAM model
- 3 A first PRAM algorithm: multiplying an array by 2
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- 5 Search on an EREW PRAM, and broadcast
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- 8 Prefix computation on a list**
- 9 Evaluating the performance of PRAM algorithms
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Prefix computation on a list (cont.)

Given a linked list of values $(x_1, \dots, x_N)$ and a binary associative operation $\otimes$, compute the sequence $(y_1, \dots, y_N)$ such that $y_k = x_1 \otimes \dots \otimes x_k$.

- Use the pointer jumping technique.
 - Each element i applies itself to its $next[i]$ before jumping.
 - $O(\log N)$ iterations at most.
 - What type of PRAM is needed?

SCAN($x, next$)

```
1: forall  $i = 1, \dots, N$  in parallel do  
2:    $y[i] \leftarrow x[i]$   
3:   while there exists a node  $i$  such that  
    $next[i] \neq \text{Nil}$  do  
4:     forall  $i = 1, \dots, N$  in parallel do  
5:       if  $next[i] \neq \text{Nil}$  then  
6:          $y[next[i]] \leftarrow y[i] \otimes y[next[i]]$   
7:          $next[i] \leftarrow next[next[i]]$ 
```

From prefix computation to list ranking

Can the prefix computation on a list be used to compute list ranking?

- Yes! Take $x[i] = 1$ for every node and $\otimes = +$:
 $y[i]$ = number of nodes from the head of the list up to i included, the **rank from the head**. The distance to the end is then $d[i] = N - y[i]$.
- Complexity: lines 1–2 and 4–5 in $O(1)$, line 3 in $O(\log N)$ (SCAN): $T(N) = O(\log N)$ with N PUs, same as RANKCOMPUTATION, on the same PRAM model.
- SCAN is a **fundamental algorithmic block** of parallel computing, usable for many things, just like BCAST and REDUCE seen earlier.
- Without subtraction: reverse the list first ($prev[next[i]] \leftarrow i$, $O(1)$, exclusive writes since $next$ is injective), take $x = 0$ for the last node and 1 elsewhere, and run the prefix computation from the tail: $y[i] = d[i]$ directly.

RANKBYPREFIX($next[1 \dots N]$)

```
1: forall  $i = 1, \dots, N$  in parallel do
2:    $x[i] \leftarrow 1$ 
3:  $y[1 \dots N] \leftarrow \text{SCAN}(x, next)$ 
4: forall  $i = 1, \dots, N$  in parallel do
5:    $d[i] \leftarrow N - y[i]$ 
6: return  $d[1 \dots N]$ 
```

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Evaluating the performance of PRAM algorithms

For a problem $A(n)$ of size n and corresponding sequential and parallel algorithms, we define the following metrics:

- $T_{seq}(n)$: execution time of the best sequential algorithm
- $T(n, p)$: parallel execution time using p PUs
- $C(n, p) = p T(n, p)$: parallel execution cost using p PUs
- $W(n, p)$: total work performed in parallel (total number of operations across all PUs)

Evaluating the performance of PRAM algorithms (cont.)

- When the algorithm uses the maximum number of PUs for the problem (e.g., $p = n$), for simplicity we just write $W(n) = W(n, p)$ (same for $T(n)$ and $C(n)$).
- $D(n) = \lim_{p \rightarrow \infty} T(n, p)$: **depth** of the parallel algorithm
- $T_{seq}(n) \leq W(n, p) \leq C(n, p)$: parallelism potentially adds an overhead
- $S(n, p) = \frac{T_{seq}(n)}{T(n, p)}$: **speedup**
- $E(n, p) = \frac{S(n, p)}{p} = \frac{T_{seq}(n)}{p T(n, p)} = \frac{T_{seq}(n)}{C(n, p)}$: **efficiency**
- If $W(n, p) = O(T_{seq}(n))$, the algorithm is called **work-optimal**.
- If $C(n, p) = O(T_{seq}(n))$, the algorithm is called **efficient**.

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Work and cost: MULTIPLYBYTwo

We now evaluate every algorithm of this session with N PUs: $C(N) = N \cdot T(N)$, and $W(N)$ counts the operations of the *active* PUs only. **Convention:** a PU whose **if** condition is false is **idle** at that step: it is allocated, hence counts in the cost, but does no work.

- $T_{seq}(N) = O(N)$: one loop over the array.
- $T(N) = O(1)$: one read, one multiplication, one write per PU, all at once;
 $C(N) = N \cdot O(1) = O(N)$.
- $W(N) = O(N)$: every PU is active during the $O(1)$ steps.
- $W(N) = O(T_{seq}(N))$: **work-optimal**;
 $C(N) = O(T_{seq}(N))$: **efficient**. $S(N) = O(N)$,
 $E(N) = O(1)$: the ideal case.

MULTIPLYBYTwo($A[1 \dots N]$)

```
1: forall  $i = 1, \dots, N$  in parallel do
2:    $A[i] \leftarrow 2 \cdot A[i]$ 
```

Model	$T_{seq}(N)$	$T(N)$	$C(N)$	$W(N)$	Work-optimal?	Efficient?
EREW	$O(N)$	$O(1)$	$O(N)$	$O(N)$	yes	yes

Work and cost: SEARCHINDEX

On a CREW PRAM (unique elements) or a CRCW PRAM (arbitrary, priority, or min/max fusion mode).

- $T_{seq}(N) = O(N)$: scan the array.
- $T(N) = O(1)$: all comparisons at once; at most one write of idx (CREW) or a concurrent write resolved by the mode (CRCW).
- $C(N) = O(N)$; $W(N) = O(N)$: every PU does one comparison.
- **Work-optimal** and **efficient** on both models.
- The concurrent reads of e are what forbids this time on an EREW PRAM: next slides.

SEARCHINDEX($A[1 \dots N], e$)

```

1:  $idx \leftarrow 0$ 
2: forall  $i = 1, \dots, N$  in parallel do
3:   if  $A[i] = e$  then
4:      $idx \leftarrow i$ 
5: return  $idx$ 

```

Model	$T_{seq}(N)$	$T(N)$	$C(N)$	$W(N)$	Work-optimal?	Efficient?
CREW	$O(N)$	$O(1)$	$O(N)$	$O(N)$	yes	yes
CRCW	$O(N)$	$O(1)$	$O(N)$	$O(N)$	yes	yes

Work and cost: BCAST

Broadcast of one value into $pe[1 \dots N]$ on an EREW PRAM, by doubling.

- $T_{seq}(N) = O(N)$: write N copies one by one.
- $T(N) = O(\log N)$: $\lceil \log_2 N \rceil$ iterations of an $O(1)$ body; $C(N) = O(N \log N)$.
- $W(N)$: at iteration k , only the PUs with $i \leq 2^k$ are active: $W(N) \leq 1 + 2 + 4 + \dots < 2N = O(N)$.
- $W(N) = O(T_{seq}(N))$: **work-optimal**.
 $C(N) \neq O(T_{seq}(N))$: **not efficient**: most PUs are idle most of the time, but the cost pays for them anyway. $S(N) = O(N/\log N)$,
 $E(N) = O(1/\log N)$.

BCAST(e, N)

```

1:  $pe[1] \leftarrow e$ 
2: for  $k = 0, \dots, \lceil \log_2 N \rceil - 1$  do
3:   forall  $i = 1, \dots, N$  in parallel do
4:     if  $i \leq 2^k$  and  $i + 2^k \leq N$  then
5:        $pe[i + 2^k] \leftarrow pe[i]$ 
6: return  $pe[1 \dots N]$ 

```

Model	$T_{seq}(N)$	$T(N)$	$C(N)$	$W(N)$	Work-optimal?	Efficient?
EREW	$O(N)$	$O(\log N)$	$O(N \log N)$	$O(N)$	yes	no

Work and cost: SEARCHINDEXEREW

BCAST of the key, then the comparisons, on an EREW PRAM.

- $T_{seq}(N) = O(N)$.
- $T(N) = O(\log N) + O(1) = O(\log N)$: the broadcast dominates; $C(N) = O(N \log N)$.
- $W(N) = O(N)$ (BCAST) + $O(N)$ (one comparison per PU) = $O(N)$.
- **Work-optimal, not efficient**: the profile of BCAST, since the comparison adds only $O(1)$ time and $O(N)$ work.
- Versus CREW: same work, $\log N$ times more time and cost. The price of exclusive reads is paid in *time*, not in work.

SEARCHINDEXEREW($A[1 \dots N], e$)

```

1:  $idx \leftarrow 0$ 
2:  $pe[1 \dots N] \leftarrow \text{BCAST}(e, N)$ 
3: forall  $i = 1, \dots, N$  in parallel do
4:   if  $A[i] = pe[i]$  then
5:      $idx \leftarrow i$ 
6: return  $idx$ 

```

Model	$T_{seq}(N)$	$T(N)$	$C(N)$	$W(N)$	Work-optimal?	Efficient?
EREW	$O(N)$	$O(\log N)$	$O(N \log N)$	$O(N)$	yes	no

Work and cost: SUMARRAYCRCW

Sum of an array on a CRCW PRAM in fusion mode +.

- $T_{seq}(N) = O(N)$.
- $T(N) = O(1)$: one read and one concurrent write per PU; $C(N) = O(N)$, $W(N) = O(N)$.
- **Work-optimal** and **efficient**, with speedup $O(N)$.
- Too good to be true: the $N - 1$ additions are done by the memory during the concurrent write, and are not counted. This is the CRCW model being unrealistic, not a good algorithm.

SUMARRAYCRCW($A[1 \dots N]$)

```

1:  $sum \leftarrow 0$ 
2: forall  $i = 1, \dots, N$  in parallel do
3:    $sum \leftarrow A[i]$            ► fusion +
4: return  $sum$ 

```

Model	$T_{seq}(N)$	$T(N)$	$C(N)$	$W(N)$	Work-optimal?	Efficient?
CRCW (fusion +)	$O(N)$	$O(1)$	$O(N)$	$O(N)$	yes	yes

Work and cost: SUMARRAYEREW and REDUCE

Sum/reduction of an array by halving, on a CREW or EREW PRAM (same analysis for both).

- $T_{seq}(N) = O(N)$: $N - 1$ applications of $\otimes$.
- $T(N) = O(\log N)$: $\lceil \log_2 N \rceil$ levels of an $O(1)$ body; $C(N) = O(N \log N)$ with N PUs (half of them never active!).
- $W(N)$: $\lceil N/2 \rceil$ active PUs at the first level, then $\lceil N/4 \rceil, \dots$: $W(N) = \lceil N/2 \rceil + \lceil N/4 \rceil + \dots \leq N + \lceil \log_2 N \rceil = O(N)$.
- **Work-optimal, not efficient**: the mirror image of BCAST, with the same profile.
 $S(N) = O(N/\log N)$, $E(N) = O(1/\log N)$. Can we recover efficiency with fewer PUs? Part 2 (Brent's theorem).

REDUCE($A[1 \dots n]$, $\otimes$)

```

1: if  $n = 1$  then return  $A[1]$ 
2: forall  $i = 1, \dots, \lceil n/2 \rceil$  in parallel do
3:   if  $2i \leq n$  then
4:      $A[i] \leftarrow A[2i - 1] \otimes A[2i]$ 
5:   else
6:      $A[i] \leftarrow A[2i - 1]$ 
7: return REDUCE( $A[1 \dots \lceil n/2 \rceil]$ ,  $\otimes$ )

```

Model	$T_{seq}(N)$	$T(N)$	$C(N)$	$W(N)$	Work-optimal?	Efficient?
CREW/EREW	$O(N)$	$O(\log N)$	$O(N \log N)$	$O(N)$	yes	no

Work and cost: RANKCOMPUTATION (list ranking)

- $T_{seq}(N) = O(N)$. Parallel, with N PUs: $\lceil \log_2 N \rceil$ iterations of an $O(1)$ body, plus the termination test of line 3, paid according to the model: $O(1)$ per iteration on CRCW, $T(N) = O(\log N)$; a reduction per test on CREW (plus a broadcast on EREW): $T(N) = O(\log^2 N)$ if tested at every iteration, $O(\log N)$ with the amortized test*.
- $W(N)$: node i stays active for $\lceil \log_2(d[i] + 1) \rceil$ iterations, most nodes being far from the end: $W(N) = O(N \log N)$, $C(N) = O(N \log N)$.
- **Not work-optimal, not efficient**: $\log N$ times more work than the sequential walk.

RANKCOMPUTATION($next[1 \dots N]$)

```

1: forall  $i = 1, \dots, N$  in parallel do
2:   if  $next[i] = \text{Nil}$  then  $d[i] \leftarrow 0$ 
   else 1
3: while  $\exists i$  with  $next[i] \neq \text{Nil}$  do
4:   forall  $i = 1, \dots, N$  in parallel do
5:     if  $next[i] \neq \text{Nil}$  then
6:        $d[i] \leftarrow d[i] + d[next[i]]$ 
7:        $next[i] \leftarrow next[next[i]]$ 

```

Model	$T_{seq}(N)$	$T(N)$	$C(N)$	$W(N)$	Work-optimal?	Efficient?
CRCW	$O(N)$	$O(\log N)$	$O(N \log N)$	$O(N \log N)$	no	no
CREW/EREW*	$O(N)$	$O(\log N)$	$O(N \log N)$	$O(N \log N)$	no	no

* termination test every $\lceil \log_2 N \rceil$ iterations (amortized); an extra $\log N$ factor on $T(N)$ and $C(N)$ otherwise.

Work and cost: SCAN

Prefix computation on a list by pointer jumping: same loop structure as RANKCOMPUTATION.

- $T_{seq}(N) = O(N)$: one walk along the list.
- Same loop as RANKCOMPUTATION: $\lceil \log_2 N \rceil$ iterations, same termination test, same active PUs at each iteration.
- $T(N) = O(\log N)$ (CRCW; CREW/EREW with the amortized test*), $C(N) = O(N \log N)$, $W(N) = O(N \log N)$.
- **Not work-optimal, not efficient**, on every model: pointer jumping trades a $\log N$ factor of work for an exponential gain in time.

SCAN($x, next$)

```

1: forall  $i = 1, \dots, N$  in parallel do
2:    $y[i] \leftarrow x[i]$ 
3:   while  $\exists i$  with  $next[i] \neq \text{Nil}$  do
4:     forall  $i = 1, \dots, N$  in parallel do
5:       if  $next[i] \neq \text{Nil}$  then
6:          $y[next[i]] \leftarrow y[i] \otimes y[next[i]]$ 
7:          $next[i] \leftarrow next[next[i]]$ 

```

Model	$T_{seq}(N)$	$T(N)$	$C(N)$	$W(N)$	Work-optimal?	Efficient?
CRCW	$O(N)$	$O(\log N)$	$O(N \log N)$	$O(N \log N)$	no	no
CREW/EREW*	$O(N)$	$O(\log N)$	$O(N \log N)$	$O(N \log N)$	no	no

* termination test every $\lceil \log_2 N \rceil$ iterations (amortized); an extra $\log N$ factor on $T(N)$ and $C(N)$ otherwise.

Work and cost: RANKBYPREFIX

List ranking derived from the prefix computation.

- $T_{seq}(N) = O(N)$.
- Lines 1–2 and 4–5: $O(1)$ time and $O(N)$ work with N PUs. Line 3: SCAN, $O(\log N)$ time and $O(N \log N)$ work.
- $T(N) = O(\log N)$, $C(N) = O(N \log N)$, $W(N) = O(N \log N)$: the profile of SCAN, on the same models.
- **Not work-optimal, not efficient.** Work-optimal list ranking exists, but is much harder (not covered).

RANKBYPREFIX(*next*[1...*N*])

```

1: forall  $i = 1, \dots, N$  in parallel do
2:    $x[i] \leftarrow 1$ 
3:    $y \leftarrow \text{SCAN}(x, \text{next})$ 
4:   forall  $i = 1, \dots, N$  in parallel do
5:      $d[i] \leftarrow N - y[i]$ 
6:   return  $d[1 \dots N]$ 

```

Model	$T_{seq}(N)$	$T(N)$	$C(N)$	$W(N)$	Work-optimal?	Efficient?
CRCW	$O(N)$	$O(\log N)$	$O(N \log N)$	$O(N \log N)$	no	no
CREW/EREW*	$O(N)$	$O(\log N)$	$O(N \log N)$	$O(N \log N)$	no	no

* termination test every $\lceil \log_2 N \rceil$ iterations (amortized); an extra $\log N$ factor on $T(N)$ and $C(N)$ otherwise.

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